- *Infrastructure as a Service (IaaS)*—Offers fundamental resources like computing, storage, and networking capabilities, using virtual machines such as Amazon EC2, Google Compute Engine, and Microsoft Azure Virtual Machines.
- *Platform as a Service (PaaS)*—Provides platforms to deploy custom applications to the cloud, such as AWS Lambda, AWS App Runner, Google App Engine, and Heroku.
- *Software as a Service (SaaS)*—Combines infrastructure and software running in the cloud, including office applications like Amazon WorkSpaces, Google WorkSpace, and Microsoft 365.

AWS is a cloud-computing provider with a wide variety of IaaS, PaaS, and SaaS offerings. Let's go into a bit more detail about what AWS is and does.

1.1 What is Amazon Web Services (AWS)?

Amazon Web Services (AWS) is a platform of web services that offers solutions for computing, storing, and networking, at different layers of abstraction. For example, you can attach volumes to a virtual machine—a low level of abstraction—or store and retrieve data via a REST API—a high level of abstraction. Use the services provided by AWS to host websites, run enterprise applications, and mine tremendous amounts of data. *Web services* are accessible via the internet by using typical web protocols (such as HTTP) and are used by machines or by humans through a UI. The most prominent services provided by AWS are *EC2*, which offers virtual machines, and *S3*, which offers storage capacity. Services on AWS work well